

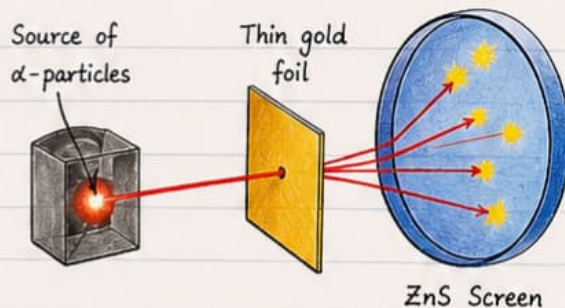
NUCLEI

The **Nucleus** is the tiny, dense and positively charged **central part** of an atom.

1. DISCOVERY

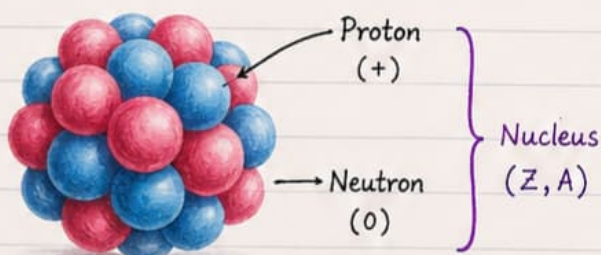
- 1911 → Rutherford's α -particle scattering experiment.
- He observed that most α -particles passed through the gold foil, but a few were deflected at large angles.
- **Conclusion** → Atom has a small, heavy, positively charged nucleus.

RUTHERFORD'S EXPERIMENT



2. NUCLEAR SIZE & CHARGE

- Radius of nucleus, $R = R_0 A^{1/3}$
where, $R_0 = 1.2 \times 10^{-15} \text{m}$
 A = mass number
- Charge on nucleus, $Ze = +Ze$
where, Z = atomic number
 e = electronic charge



$$A = Z + N$$

(Mass-number) (Protons) (Neutrons)

3. NUCLEAR COMPOSITION

- Nuclei are made up of protons and neutrons called **nucleons**.
- **Proton** → positive charge (+e)
- **Neutron** → no charge (0)
- Mass of proton $\approx$ Mass of neutron $\approx 1u$
- Mass of electron is negligible.

Particle	Charge	Mass (u)
Proton (p)	+e	1.0073
Neutron (n)	0	1.0087
Electron (e)	-e	0.00055

4. MASS DEFECT (Δm)

- Actual mass of nucleus is less than the sum of masses of individual nucleons.
 $\Delta m = Zm_p + Nm_n - M_{\text{nucleus}}$
- This missing mass is called **mass defect**.
- It is converted into energy which holds the nucleus together.

5. BINDING ENERGY (B.E.)

- Energy required to break the nucleus into its individual nucleons.
 $B.E. = \Delta m \times c^2$
- More the binding energy → more stable is the nucleus.

★ KEY POINT

Nuclei are tiny but powerful.
They hold the secret of immense energy.

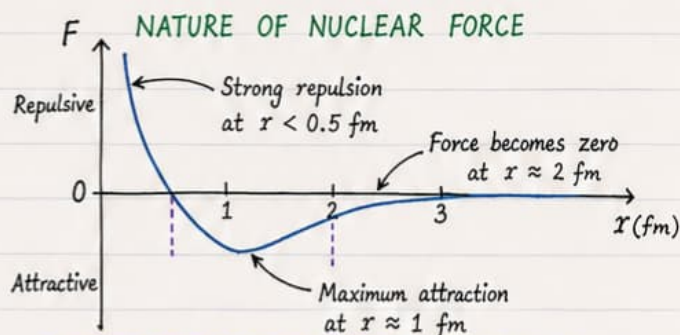


6. NUCLEAR FORCE

- Strong force of attraction between nucleons is called **nuclear force**.
- It is **charge independent** and short range force.
- It is **much stronger** than electrostatic repulsive force between protons.

$$\text{Range} \approx 1 - 2 \text{ fm } (10^{-15} \text{ m})$$

(fm = femtometer)

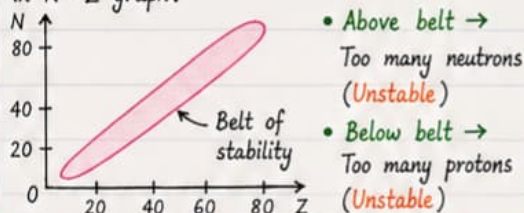


Nuclear force → Very strong (short range)
 → Repulsive at $r < 0.5 \text{ fm}$
 → Attractive at $0.5 \text{ fm} < r < 2 \text{ fm}$
 → Zero at $r \approx 2 \text{ fm}$

7. STABILITY OF NUCLEI

- Nuclei are held together by nuclear force.
- For small nuclei, $N = Z$ (no. of neutrons = no. of protons).
- For heavier nuclei, $N > Z$ (extra neutrons provide stability).

STABLE NUCLEI lie in the belt shown in $N-Z$ graph.



8. RADIOACTIVITY

- Nuclei of some atoms are unstable.
- They decay spontaneously by emitting radiations. This phenomenon is called **radioactivity**.
- Discovered by → **Henri Becquerel (1896)**
- Types of radioactive radiations

Radiation	Symbol	Nature	Charge	Penetrating Power	Speed
Alpha	α	Helium nucleus	+2e	Low	$\sim 10^7 \text{ m/s}$
Beta	β	Electron	-e	Medium	$\sim 10^7 \text{ m/s}$ ($\approx c$)
Gamma	γ	Electro-magnetic	0	Very High	c (Light speed)

9. RADIOACTIVE DECAY LAW

- The rate of disintegration is proportional to the number of undecayed nuclei present.
- Mathematical form →

$$-\frac{dN}{dt} = \lambda N$$

where, N = number of undecayed nuclei at time t
 λ = decay constant (s^{-1})
 t = time

Half-life ($T_{1/2}$) = time in which half of the nuclei decay.

$$T_{1/2} = \frac{0.693}{\lambda}$$

It is independent of initial number of nuclei.

Solutions:

- $N = N_0 e^{-\lambda t}$
- $N = \frac{N_0}{2^{t/T_{1/2}}}$
- $A = A_0 e^{-\lambda t}$

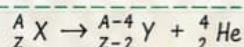
★ KEY POINT

Radioactivity is a random process, so we can only predict probability, not individual decay.



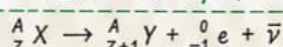
10. MODES OF RADIOACTIVE DECAY

(i) ALPHA DECAY ★



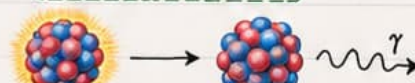
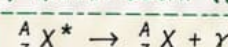
- Emission of alpha particle (${}_2^4 \text{He}$ nucleus).
- A decreases by 4, Z decreases by 2.
- Example: ${}_{92}^{238} \text{U} \rightarrow {}_{90}^{234} \text{Th} + {}_2^4 \text{He}$

(ii) BETA DECAY (β^-) ★



- Emission of electron (β^- particle).
- Z increases by 1, A remains same.
- Example: ${}_{6}^{14} \text{C} \rightarrow {}_{7}^{14} \text{N} + {}_{-1}^0 e + \bar{\nu}$

(iii) GAMMA DECAY (γ) ★



- Emission of gamma photon.
- Neither A nor Z changes.
- Example: ${}_{27}^{60} \text{Co}^* \rightarrow {}_{27}^{60} \text{Co} + \gamma$



Radioactive decay is nature's way of bringing unstable nuclei to stability.

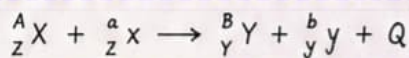


NUCLEI

STUDY ✓
SMART ✓
SCORE ✓
HIGH ★

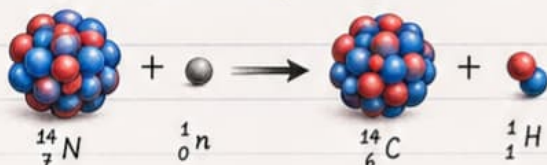
11. NUCLEAR REACTIONS

- When a bombarding particle hits the nucleus, the nucleus breaks into two parts or a **new nucleus** is formed.
- General form →



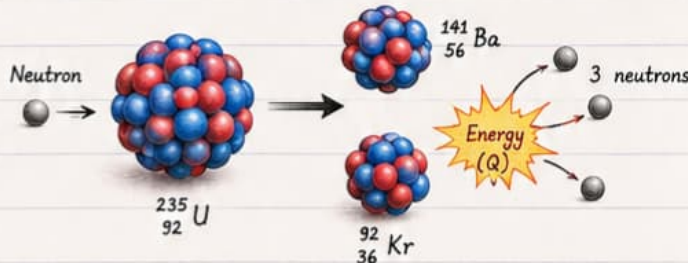
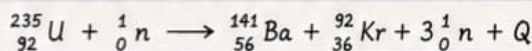
Where, Q = energy released (+) or absorbed (-) in the reaction.

- (ii) Formation of ${}^{14}_6\text{C}$ in upper atmosphere



EXAMPLE ★

- (i) Neutron induced fission of Uranium



★ Key Points

- In nuclear reactions mass and energy are conserved.
- Nuclear reactions involve very large energies (MeV range).



12. ENERGY IN NUCLEAR REACTIONS (Q-VALUE)

- The energy released or absorbed in a nuclear reaction is called **Q-value**.

$$Q = [(m_i - m_f)c^2]$$

m_i = initial total mass

m_f = final total mass

c = speed of light

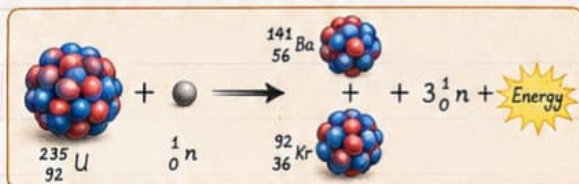
$Q > 0 \rightarrow$ Energy released (Exothermic reaction)

$Q < 0 \rightarrow$ Energy absorbed (Endothermic reaction)

13. NUCLEAR FISSION & FUSION

(A) NUCLEAR FISSION

- Splitting of a heavy nucleus into two (or more) lighter nuclei with **release of energy**.
- Example → Fission of ${}^{235}_{92}\text{U}$

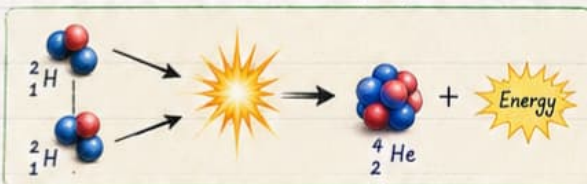


Features

- Occurs in heavy nuclei ($A > 200$)
- Chain reaction possible
- Used in nuclear reactors and atomic bombs
- Example elements: ${}^{235}_{92}\text{U}$, ${}^{239}_{94}\text{Pu}$

(B) NUCLEAR FUSION

- Combination of two light nuclei to form a heavier nucleus with **release of large energy**.
- Example → Fusion in Sun (Hydrogen to Helium)



Features

- Occurs in light nuclei ($A < 50$)
- Very large amount of energy released
- Powers the Sun and stars
- Example: ${}^2_1\text{H}$, ${}^3_1\text{H}$ (Isotopes of Hydrogen)

14. NUCLEAR BINDING ENERGY (B.E.) & BINDING ENERGY CURVE

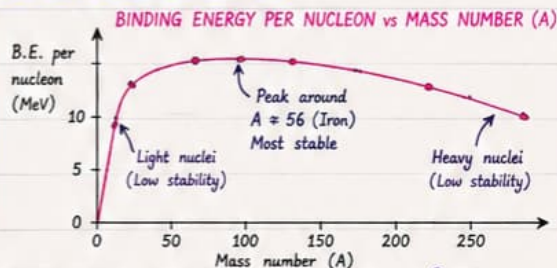
- Energy required to separate a nucleus into its individual nucleons.

- For a nucleus,

$$\text{B.E.} = [Zm_p + (A - Z)m_n - M_{\text{nucleus}}]c^2$$

- Binding energy per nucleon

$$= \frac{\text{B.E.}}{A}$$



★ Key idea

- Energy is released in fusion of light nuclei.
- Energy is released in fission of heavy nuclei.

Remember: Nuclei with higher binding energy per nucleon are more stable.



CHAPTER-13 NUCLEI

STUDY ✓
SMART ✓
SCORE ✓
HIGH ★

15. NUCLEAR DENSITY

- Nuclear radius, $R \propto A^{1/3}$
- Volume, $V \propto R^3 \propto A$
- Nuclear mass, $M \propto A$
- Hence, density = Mass/Volume

$$\rho = \frac{M}{V} \propto \frac{A}{A} = \text{constant}$$

Nuclear density is constant and
 $\approx 2.3 \times 10^{17} \text{ kg m}^{-3}$

VISUALIZING NUCLEAR SIZE



$$R \approx 1.2 A^{1/3} \text{ fm}$$

(1 fm = 1 femtometer = 10^{-15} m)

16. NUCLEAR MODELS

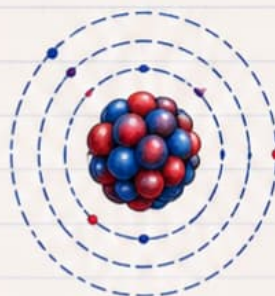
(i) LIQUID DROP MODEL

- Nucleus behaves like a drop of incompressible liquid.
- Explains fission, fusion and nuclear binding energy.
- Limitations:** Cannot explain magic numbers and spin.



(ii) SHELL MODEL

- Nucleons move in shells like electrons in an atom.
- Explains magic numbers (2, 8, 20, 28, 50, 82, 126) and spin.
- Based on Pauli exclusion principle.



Magic Numbers

2, 8, 20,
28, 50,
82, 126

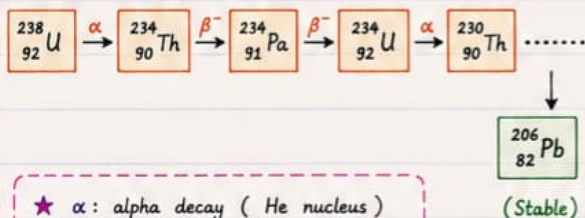
17. APPLICATIONS OF NUCLEAR ENERGY

Application	Description	Examples / Use	
Power generation	Nuclear fission in reactors produces huge amount of energy.	Nuclear power plants (Tarapur, Kudankulam)	
Medical uses	Radioisotopes are used in diagnosis and treatment.	Iodine-131 (thyroid), Cobalt-60 (cancer therapy)	
Agriculture	Improves crop yield, controls pests and preserves food.	Mutation breeding, food irradiation	
Research	Helps in studying structure of matter and fundamental particles.	Particle accelerators, radioactive tracers	

18. RADIOACTIVE SERIES

- Natural decay sequences of heavy elements.

Example: Uranium series



★ α : alpha decay (He nucleus)
 β^- : beta minus decay (electron⁺)

QUICK REVISION

- Nucleus** → tiny, dense, +ve charged central part.
- Composition** → protons (+) & neutrons (0).
- Force** → nuclear force (strong, short range).
- Stability** → $N = Z$ for light nuclei,
 $N > Z$ for heavy nuclei.
- Binding Energy** → Energy needed to break nucleus.
- Radioactivity** → spontaneous emission of radiations.
- Fission** → heavy nucleus → lighter nuclei.
- Fusion** → light nuclei → heavy nucleus.
- Density of nucleus $\approx 2.3 \times 10^{17} \text{ kg m}^{-3}$

★ **KEY TAKEAWAY** Nuclei may be tiny, but they hold enormous energy.

Understanding them unlocks the power of the universe!

